$$I = \bar{I} - bi, b > 0$$

Investment (I) contains an autonomous part and is inversely related to the interest rate (i) as it is the cost of borrowing more capital. b is the responsiveness of investment to the interest rate. Higher b implies that even with a smaller change in interest rate, the investment will change by a greater magnitude. As interest rate goes up, investment in the economy declines by the unit 'b'.

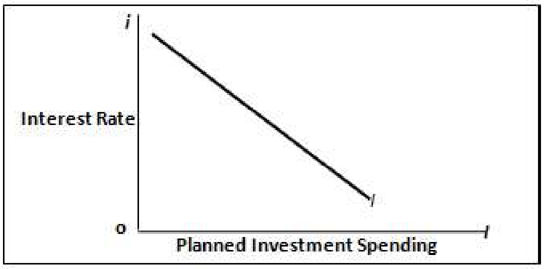


Figure 11.3: The investment schedule

Source: Dornbuschet. al., Macroeconomics, 11th ed.

The Goods Market Equilibrium: The IS Curve

Describing the real sector of the economy, the IS curve (Fig. 11.4) represents the condition that aggregate demand (AD) equals national product (Y).

$$AD = C + G + I$$

such that:

$$C = \bar{C} + c(1-t)Y;\ I = \bar{I} - bi;\ G = \bar{G}$$

$$AD = \bar{A} + c(1-t)Y - bi$$

where, $\bar{A} = \bar{C} + \bar{I} + \bar{G}$

In equilibrium, $Y = AD$

This implies $Y = AD = \bar{A} + c(1-t)Y - bi$

$$Y - c(1-t)Y = \bar{A} - bi$$

$$Y[1 - c(1-t)] = \bar{A} - bi$$

$$Y = \alpha_g(\bar{A} - bi) \text{ where, } \alpha_g = \frac{1}{[1 - c(1-t)]}$$

Or $$i = \frac{\bar{A}}{b} - \frac{Y}{\alpha_g b} \qquad (3)$$

α_g can be understood as the multiplier. The slope of the IS curve is determined by the value of the multiplier and responsiveness of investment to the interest rate (b).

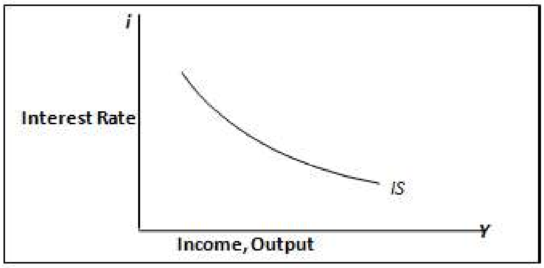


Fig. 11.4: The IS schedule

The Money Market Equilibrium: LM Curve

The demand for real money balance (L) is a positive function of the real income and a negative function of the real interest rate (Figure 11.5). It is given by:

$L = kY - hi$, such that $k,\ h > 0$

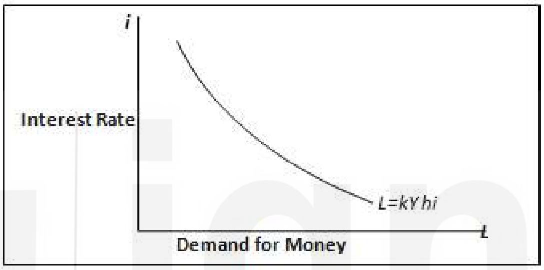


Fig. 11.5: The demand for real balances schedule

Source: Dornbuschet. al., Macroeconomics, 11[th]ed.

The money market equilibrium: Describing the money market of the economy, the LM curve (Figure 11.6) represents the condition that demand for real balances equal real money supply. The nominal stock of money M is determined by the Central Bank, the Reserve Bank of India (RBI) for India. We take the nominal stock of money as given at $\overline{M}$. We also assume prices to be constant at $\overline{P}$, hence the real money supply is $\frac{\overline{M}}{\overline{P}}$. At equilibrium, demand for real balances = supply of real balances, that is,

$$L = \frac{\overline{M}}{\overline{P}}$$

This implies, $kY - hi = \frac{\overline{M}}{\overline{P}}$

$$i = \frac{1}{h}\left[kY - \frac{\overline{M}}{\overline{P}}\right] \qquad (4)$$

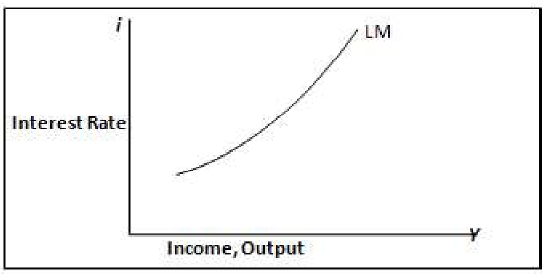


Fig.11.6: The LM Schedule

Source: Dornbuschet. al., Macroeconomics, 11[th]ed.

The equilibrium interest rate in the economy is determined by the interaction of the goods and money market (through Equations 3 and 4). The equilibrium level of income is hence gets simultaneously determined.

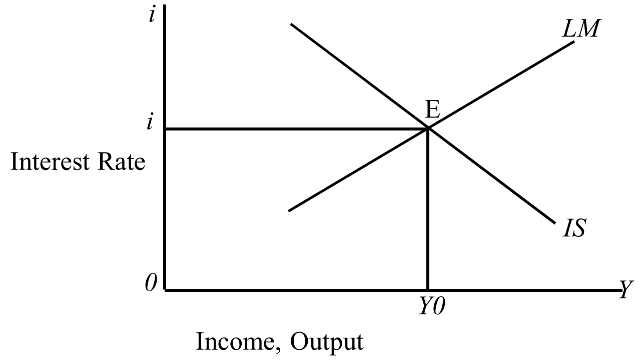


Fig. 11.7: The IS-LM and determination of equilibrium in the economy

Source: Dornbuschet. al., Macroeconomics, 11th ed.

11.2.1 Implications of IS-LM Framework for Fiscal Policy

Change in government spending ($\Delta\overline{G}$): If there is an *increase* in the government spending such that the increase in spending is expressed as $\Delta\overline{G}$, this shifts the IS curve outwards to the right. At this stage there is no change in LM curve. The new equilibrium income and interest rate level is determined where the new IS curve intersects the LM curve (E'). The change in the equilibrium level of income will be:

$$\Delta Y = \alpha_g \Delta\overline{G}$$

The new interest rate will be higher than the original interest rate and the change in Y is much more than the initial change in G.

Mechanics: Recalling what has been studies earlier, the increase in autonomous government spending increases the aggregate demand (AD) in the economy shifting IS curve to IS'. This raises the output to Y". The increase in income increases the demand for real balances in the economy. Given that the money supply is fixed by the RBI, interest rate has to rise (from i to i' so that money market reaches equilibrium. This increase in interest rate decreases the investment and hence a decline in the overall AD and output also declines to Y'. This decline in private investment as a result of increasing government spending (increase in public investment) is also known as *crowding-out effect* of public investment. Since government spending has to be financed, usually from increased government borrowings, it leaves less of the country's savings to be borrowed by the private sector and that also raises the cost of borrowing for the private sector, leading to crowding-out of private investments. However, evidence from India suggests otherwise, as there exist infrastructural bottlenecks. Hence increasing the public investment actually crowds-in the private investment. Between 1971-71 to 2002-03, there has not been evidence of crowding-out of the private investment due to public investment, rather complementarity is observed between the two[1].

[1]Lekha S. Chakraborty; Fiscal Deficit, capital formation and crowding out: Evidence from India.

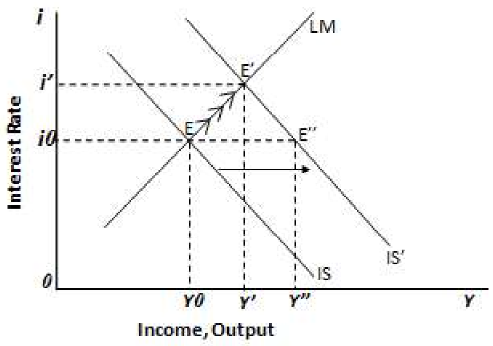


Fig.11.8: Increase in government spending

Source: Dornbuschet. al., Macroeconomics, 11[th]ed.

Increase in government spending is an expansionary fiscal policy. In case of a contractionary fiscal policy measure, the government can reduce the autonomous government spending, shifting the IS curve leftward and reducing the equilibrium level of interest rate and income.

Change in taxes (t): A change in the proportional income tax rate (t) is not as straight forward. Since the change in the income tax affects the disposable income, it changes the consumption. Let us consider an increase in tax rate from *t* to *t'* (for e.g., increase in personal income tax from 10 per cent to 20 per cent). This increase in taxes clearly reduces the consumption expenditure by the household as they now have less income at their disposal. A change in the tax rate is going to affect the slope of the IS curve as it changes the multiplier. With taxes being increased to t' (contractionary fiscal policy), the slope of the IS curve will increase and the overall value of the multiplier will decrease. A reduction in the value of multiplier implies that income/GDP will not change as much with the changes in any of its components.

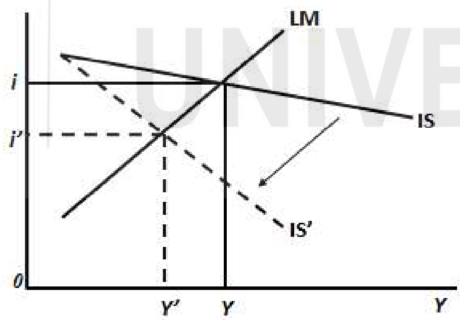


Fig. 11.9: Increase in proportional income tax

Source: Author

Mechanics: As we can see in Figure. 11.9, the slope has increased and the IS has shifted to IS'. There will be no change in LM as there is no change in the money market. With the shift in IS, the new equilibrium income and interest rate is achieved at Y' and i'. As the tax rate has gone up, the disposable income in the hand of consumers decrease. This lowers the consumption demand (C) and decreases the overall aggregate demand (AD). Since the AD decreases the equilibrium income in the economy, the demand for real balances also decrease, but as earlier the supply is fixed. The interest rate reduces as a consequence. This reduction in interest rate increases the investment demand but not by as much to offset the earlier decline in AD.

Check Your Progress 1

1) Fill in the blanks.

   i) In the Indian context, the consumption expenditure incurred by .... constitutes the majority of GDP.

   ii) The IS schedule shows the .... equilibrium.

   iii) The decline in private investment as a result of increasing government spending is known as ....

2) In terms of the IS-LM framework, explain the mechanics of increasing government spending on a closed economy which is initially in equilibrium.

   ....

   ....

   ....

   ....

   ....

11.3 CONCEPT OF FISCAL POLICY

We define *fiscal policy* to encompass any decision to change the composition of or timing of government expenditures or to vary the burden, structure, or frequency of tax payment. Governments all around the world use fiscal policy tools as counter cyclical measure i.e., to stabilise economy from cycles of boom and bust.

Fiscal policy aims at using its three major instruments– taxes, spending and borrowing– as balancing factors in the development of the economy. According to Arthor Smithies, fiscal policy is a policy under which government uses its expenditure and revenue programme to produce desirable effects and avoid undesirable effects on the national income, production and employment. Fiscal policy consists of the use of taxes, government spending and public debt operations to influence the economic activities of the community in desired ways and is concerned with the allocation of resources between the public and private sectors and their use for the attainment of stability and growth. Although the effects of fiscal policy are extensive, they are particularly measurable in areas such as employment, price stability, savings and investment, and the balance of payment. The prime aim of such a policy is to maintain a high level of employment without inflation.

11.3.1 Objectives of Fiscal Policy

Formulation of fiscal policy presumes the identification and clear recognition of the institutional aspects of government finance, such as tax system, tax incidence and shifting, budget formulation and execution and financial management. The focus of budgeting is on the attainment of efficiency in the allocation of resources within the public sector and is influenced at each stage by the goals of the fiscal policy. The changes in government income or expenditure have been designed to affect the level of activity in the economy as a whole. An understanding of the fiscal policy is essential for gaining proper perspectives on the different aspects of budgeting. In the recent years, importance of fiscal policy has increased due to economic fluctuations.

The budgetary fiscal policy can play a key role in the process of economic development by (i) mobilising additional resources, (ii) maintaining economic stability, (iii) allocating resources into socially necessary lines of development, (iv) reducing extreme inequality in income and wealth, and (v) providing the necessary incentive to the private sector for its healthy growth. The fiscal policy thus has not one goal. It has a multiplicity of goals. Hence, they cannot be achieved by any one set of policies. There may be conflicts between some of these other goals and stabilisation goals. In the context of economic growth, the fiscal policy has to be so framed as to avoid an inflationary pressure in the economy.

11.3.2 Implications of Fiscal Policy

Fiscal policy is so wide-ranging that selection of a combination of differing objectives is both complex and controversial. The decisions that the government makes over fiscal policy can have important implications for business. In today's globalised economy, tax policy is not conducted in a vacuum. Taxes can influence the choices of both business and labour. Supply-side economists argue that economic activity is quite sensitive to changes in tax rates, particularly the highest marginal rates. What causes governments to ignore budget deficits in some circumstances and not in other? Business also needs to understand the effect a change in fiscal policy is likely to have on aggregate demand. In recent decades, the ratio of government expenditure to GDP has increased in most countries. In many countries its present level cannot be financed by ordinary sources of revenue. As a consequence, many governments have been compelled to borrow heavily. If a country borrows too much money, it has to pay a great deal of interest every year in order to service that debt.

The greatest obstacle to proper use of fiscal policy is that changes in fiscal policy may be necessarily bundled with other changes that please or displease various stakeholders. This is true for a tax cut for some favoured constituency. The problem of making good fiscal policy in the face of such obstacles is, in the final analysis, not economic but political. There are many practical limitations or drawbacks to the actual working of the fiscal policy. The objective of fiscal policy in modern society cannot be promoted in isolation. It has to be coordinated with monetary, credit and debt policies to be effective. The questions relating to the manner of financing deficit or disposing of surplus in a budget have close relationship with monetary and credit policies.

Monetary and fiscal policies can be best understood in the context of the events that shape them. Such an analysis can assist in choosing policies that improve rather than disrupt short- and long-term economic performance. Of particularly interest are the circumstances and policies surrounding the recent recession.

11.4 FISCAL POLICY IN INIDA

11.4.1 India's Fiscal Policy Architecture

India adopted a federal form of governance in post-independence period. The Constitution of India (Article 1) states that India, that is Bharat, shall be a Union of States. The Indian Constitution provides the overarching framework for country's fiscal policy with the assignment of taxation powers and expenditure responsibilities being divided between the Union and State governments. The Seventh Schedule of the Constitution of India specifies the legislative domain of the Union and States Governments in terms of three lists namely, (a) Union List (or List I), (b) State List (or List II), and (c) Concurrent List (or List III). These lists represent the powers conferred upon the Union Government, State Governments, and shared powers between the two respectively. All residuary powers are vested with the Union. The Constitution also provides that for every financial year, the government shall place before the legislature a statement of its proposed taxation and expenditure provisions, also known as the budget, for legislative debate and approval. The Union and the State governments each have their own annual budgets. Details on this aspect has been provided in the next unit (Unit 12).

11.4.2 Fiscal Policy until 1990s

The formation of a Planning Commission, prompted by Prime Minister Nehru's commitment, was announced in the Finance Minister's Budget Speech in February, 1950. The Commission was established on 15th March, 1950 through a cabinet resolution with primary objectives of formulating medium to long term plans, advising on allocation of funds to Ministries in the Union Government through annual budget, approving the plan of each State. The resolution clearly indicated that the Commission would essentially be an advisory body which would make recommendations to the Cabinet. The Planning Commission, by virtue of its mandate to prepare Five-year Plan came to exercise allocative powers. One of the roles that it performed was to approve Plans of each of the states and allocate transfer of funds from the Union government to States, both tied and untied under the Plan.

The main fiscal impact of the planning process is the division of expenditures into plan and non-plan components. The *plan components* relate to items dealing with long-term socio-economic goals as determined by the ongoing planning process. They often relate to specific schemes and projects. Furthermore, they are usually routed through central ministries to state governments for achieving certain desired objectives. These funds are generally in addition to the assignment of central taxes as determined by the Finance Commissions. In some cases, the state governments also contribute their own funds to the schemes. *Non-plan* expenditures broadly relate to routine expenditures of the government for administration, salaries, and the like[2].

The objective of the initial five-year plans during the *1950s and 1960s* was mainly to become a self-sufficient nation and increase the growth rate of the economy through strengthening the public sector enterprises. Taxation was used as an instrument for transferring resources to the Government to enable it to undertake large-scale public investment in an effort to spur economic growth. Both direct and indirect taxes were focussed on extracting revenues from the private sector to fund the public sector. The combined centre and state tax revenue to GDP ratio increased from 6.03 per cent in 1950-51 to 15.47 per cent in 1987-88. For the central government this ratio (i.e., central taxes net of devolution to GDP ratio) was 3.43 per cent of GDP in 1950-51 with the larger share coming from

[2] The distinction between plan and non-plan government spending has been dispensed with and now the classification of government spending follows the categorisation of revenue and capital spending.

indirect taxes at 2.20 per cent of GDP and direct taxes at 1.23 per cent of GDP. Given their low direct tax base, the direct tax-GDP ratio of state governments was 0.53 per cent while their share of indirect taxes in GDP was 1.61 per cent in 1950-51. Furthermore, the fiscal policy was geared towards achieving the economic objectives of (i) promoting employment through grant of tax incentives to new investment; (ii) reducing inequality through progressive taxes on income and wealth; (iii) reducing pressure on balance of payments through increase of import duties; and (iv) stabilising prices through tax rebate in excise duties on consumption goods[3].

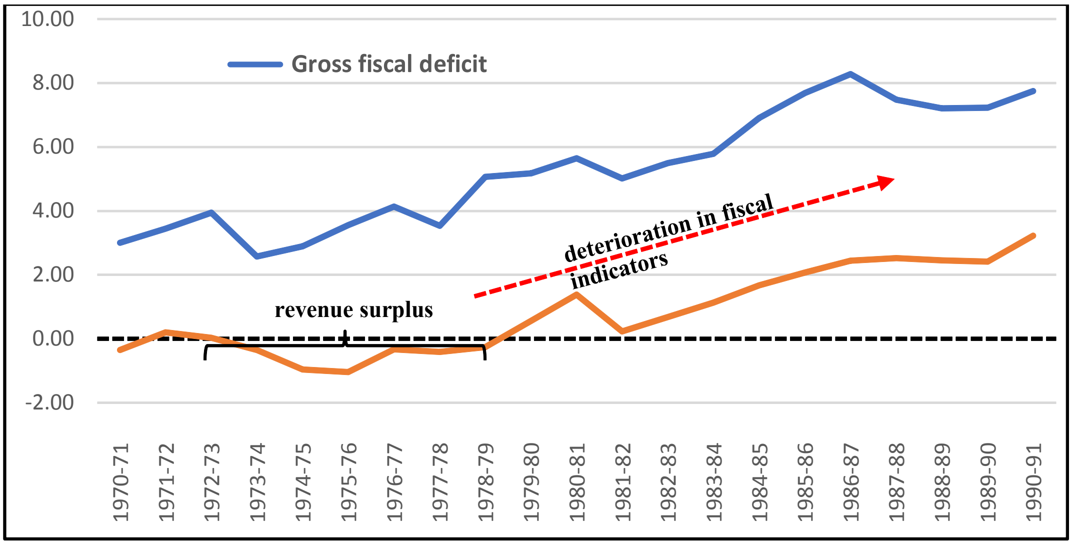
The systemic evolution of tax system in India started with the implementation of a number of recommendations of the Taxation Enquiry Commission in 1955-56 which was appointed in April, 1953. The Commission submitted its report in December, 1954. However, the government invited the British economist Nicholas Kaldor to examine the possibility of reforming the tax system. The findings and recommendations of Kaldor was published in a report in 1956. Kaldor found the system is inefficient and inequitable given the narrow tax base and inadequate reporting of property income and taxation. He also found the maximum marginal income tax rate at 92 per cent to be too high and suggested it be reduced to 45 per cent. In view of his recommendations, the government revived capital gains taxation, brought in a gift tax, a wealth tax and an expenditure tax (which was not continued due to administrative complexities).

Despite Kaldor's recommendations, for both corporate and personal income taxes the highest marginal income tax rates continued to be very high. The Direct Taxes Enquiry Committee of 1971 found that the high tax rates encouraged tax evasion and following its recommendations in 1974-75, the highest marginal rate for personal income tax was brought down to 77 per cent.

Fiscal policy during the 1970s focused on achieving greater equity and social justice and both taxation and expenditure policies were employed towards that end. Accordingly, income tax rates were raised to very high levels, with the maximum marginal rate of personal income tax moving up to 97.5 per cent including the surcharge. India's expenditure norms remained conservative till the 1980s. From 1973-74 to 1978-79 the central government continuously ran revenue surpluses (Figure 11.10). The gross fiscal deficit (GFD) also showed a slow growth with certain episodes of downward movements. Over the years, in addition to the commitment towards a large volume of developmental expenditure, the Government's expenditure widened to include rising subsidies. Large interest payments on growing debt and downward rigidity in prices further contributed to increased expenditure. Revenues, on the other hand, were less buoyant leading to the emergence of sizeable revenue deficits (RD) in the Central government budget from 1979-80 onwards.

During the decade of 1980s, the Indian public finances were in a state of disarray. We find considerable fiscal deterioration took place during the 1980s and it eventually became unsustainable, although the growth rate did rise significantly with the enhancement in public investment in infrastructure.While the revenue deficit (RD) of the central government as percentage of GDP increased from 1.4 per cent in 1980-81 to 2.44 per cent in 1989-90, its gross fiscal deficit (GFD) as percentage of GDP also increased during this period, increasing from 5.71 per cent in 1980-81 to 7.31 per cent in 1989-90.

[3] Rao and Rao, 2006

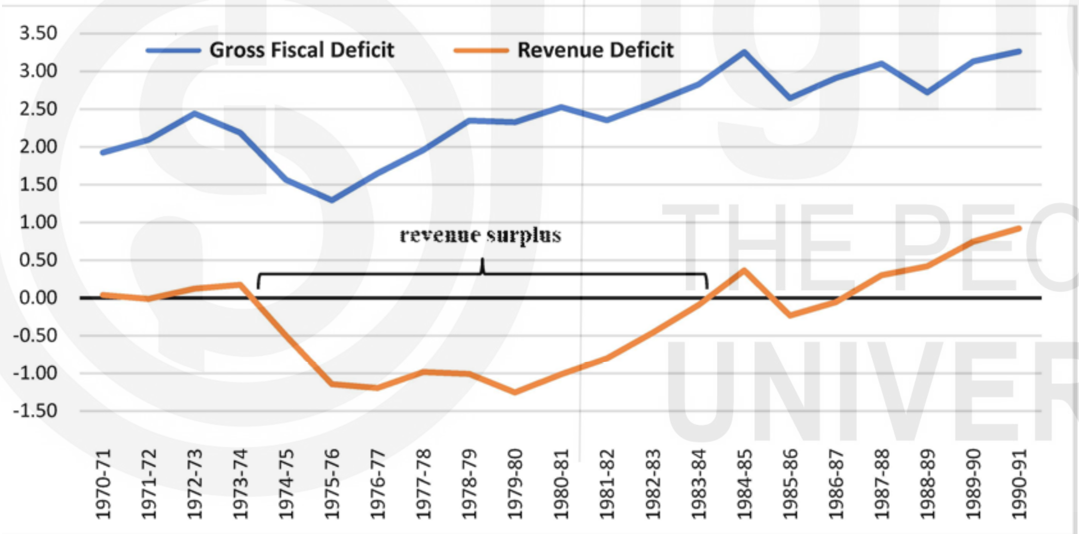


Note: Deficit (+) / Surplus (–)

Fig. 11.10: GFD and RD of the Centre (per cent of GDP)

Source: DBIE and CMIE

The *State governments* had revenue surplus during 1974-75 to 1986-87 with the exception of 1984-85 (Figure.11.11). Their fiscal position also deteriorated in the late 1980s.



Note: Deficit (+) / Surplus (–)

Fig. 11.11: GFD and RD of the States (per cent of GDP)

Source: DBIE and CMIE

Rising deficits and debt reduced the quality of the expenditure with increase in expenditure on interest payment at the cost of social and developmental expenditure. The Government sought to reduce its deficit through increase in taxes. An important reform in the indirect taxes was recommended by the Indirect Tax Enquiry in 1977. It recommended introduction of the input tax credit to convert the cascading manufacturing tax into a manufacturing value added tax (MANVAT). But this was implemented as a modified value added tax (MODVAT) in 1986 in a phased manner covering only selected commodities (Rao and Rao, 2006).

The other main central indirect tax is the customs duty. Customs duties were hiked to augment revenue and to protect domestic industry. In 1985-86 the government presented its Long-Term Fiscal Policy stressing on the need to reduce tariffs, have fewer rates and eventually remove quantitative limits on imports. Some reforms were attempted but due to revenue raising considerations the tariffs in terms of the weighted average rate increased from 38 per cent in 1980-81 to 87 per cent in 1989-90. By 1990-91 the tariff structure had a range of 0 to 400 per cent with over 10 per cent of imports subjected to tariffs of 120 per cent or more[4].

The 8th Finance Commission whose operational period was 1984-1989 made it clear that in considering competing claims on resources, the overriding consideration would be national interest. The 8th FC also made the tax devolution more progressive by leaning in favour of the backward states. It also set 5 per cent of the net proceeds of sharable excise duties exclusively for non-plan revenue deficit.

In 1970-71, direct taxes contributed to around 16 per cent of the central government's revenues, indirect taxes about 58 per cent and the remaining 26 per cent came from non-tax revenues. By 1990-91, the share of indirect taxes had increased to 65 per cent, direct taxes shrank to 13 per cent and non-tax revenues were at 22 per cent.

While the institutional arrangements in place initially appeared adequate for driving the development agenda, but the sharp deterioration of the fiscal situation in the 1980s led to the balance of payments crisis of 1991.

11.4.3 Fiscal Policy 1990s Onwards

The political uncertainties both domestic (assassination of Prime Minister Rajiv Gandhi) and abroad (The Gulf War) added to the existing fiscal deterioration of the country. Rising oil prices led to greater burden of the fuel subsidy along with increasing external debt that eventually led to a balance of payment crisis in 1991. The crisis brought about the most important economic reforms which opened up the economy to private players both domestic (dismantling of the License Raj system) and from abroad. Fiscal policy was also re-aligned with these policy changes.

To improve the falling tax receipts and to rationalise the tax structures, Tax Reform Committee was formed with the view to achieve set targets. *The Tax Reforms Committee* (TRC) concentrated on finding a suitable framework to reform both the direct and indirect tax structure in the country. The committee recommended two major reforms on direct taxes– one was the simplification and rationalisation of the direct tax structure (*The Chelliah Committee,* 1992); the other was to introduce a service tax to widen the tax base (*The Chelliah Committee,* 1994).

As a part of the subsequent *direct tax reforms*, the personal income tax brackets were reduced to three with rates of 20, 30 and 40 per cent in 1992-93. Financial assets were removed from the imposition of wealth tax and the maximum rate of wealth tax was reduced to 1 per cent. The Personal income tax rates were reduced again to 10, 20, and 30 per cent in 1997-98. These rates have largely remained the same since then with the exemption limit being increased and slab structure raised from time to time. A subsequent 2 per cent surcharge to fund education was later made applicable to all taxes.

[4] Rao and Rao, 2006